Zeyu Ni

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Awards & Certifications

Microsoft Certified: Azure Fundamentals Jun. 2026
Microsoft Certified: Azure AI Fundamentals Jun. 2026
Certificate of Distinction for the Euclid Mathematics Contest (Top 25%) Apr. 2024
National Silver for the Sir Isaac Newton Physics Contest (Top 25%) May 2024

Experience

Waterloo Experience Accelerate (WEA) Jun. 2026 – Aug. 2026
AI Researcher *Waterloo, Ontario*

- Evaluated market viability and unit economics for an early-stage vertical SaaS platform, establishing a **$600M** CAD global TAM and a **$1.6M** CAD SOM across **800** target ceramics studios.
- Analyzed AI integration strategies across competitor platforms (e.g., Fresha, GlossGenius, Wix) to design workflow-integrated AI features that reduced manual data processing time by **90%**.
- Created a data-driven **go-to-market** roadmap and hybrid pricing model aimed at improving adoption among non-technical ceramics studio owners.

Projects

Course Watcher | *Python*, *FastAPI*, *SQLAlchemy*, *OpenAI API*, *Railway*, *AI Engineering* Jun. 2026 – Present

- Built and deployed a **FastAPI** backend on **Railway** to manage course watches, retrieve live enrollment data from the University of Waterloo Open Data API, and save historical seat snapshots, using **git** for source control.
- Implemented a **token-protected** parsing endpoint using **OpenAI** structured outputs and Pydantic validation to convert English requests into normalized course, component, and term fields.
- Tested API routes, authentication, database constraints, and external-service failures using **pytest**, **HTTPX MockTransport**, and **FastAPI** dependency overrides without calling live UW or OpenAI services.

Hospital-Induced Delirium Prevention System | *Arduino*, *C++*, *Embedded Systems*, *Sensors* Sep. 2025 – Dec. 2025

- Collaborated with 3 teammates to build a portable **Arduino** system that continuously monitored room noise, light, and temperature to flag environmental risk factors associated with hospital-induced delirium.
- Implemented multi-sensor monitoring with an I2C 16×2 LCD, RGB LED indicators, and audible alerts when noise exceeded 40 dB or light exceeded 300 lux.
- Programmed an 8-sample Simple Moving Average filter and PWM adaptive lighting in **C++**, achieving zero false alarms and alert response times under 2 seconds during validation testing.

Tactical Tanks – 2D Browser Game | *JavaScript*, *HTML5*, *CSS3* Jun. 2023 – Aug. 2023

- Developed a 2D top-down tank combat browser game using vanilla **JavaScript** and **HTML5** Canvas, implementing smooth multi-directional movement, turret rotation, and projectile shooting mechanics.
- Engineered custom physics algorithms for bullet trajectory calculation and bounding-box collision detection between projectiles, tanks, and map obstacles with real-time particle explosion effects.
- Implemented menu, gameplay, and restart states with responsive keyboard controls, supporting complete play sessions across desktop browsers.

Education

University of Waterloo Sep. 2025 – Apr. 2030
Candidate for Bachelor of Computer Engineering *Waterloo, Ontario*

- **Relevant Coursework:** Data Structures and Algorithms (C++), Object-Oriented Programming, Linear Algebra, Calculus, Digital Circuit, Digital Computers

Skills

Languages: C++, C, Python, Java, JavaScript, TypeScript, SQL, Bash, HTML, CSS
Frameworks & Libraries: FastAPI, React, Node.js, Express.js, Django, Flask, Spring Boot, SQLAlchemy
Databases & APIs: PostgreSQL, MySQL, SQLite, MongoDB, REST APIs, gRPC, OpenAI API
Tools & Platforms: Git, GitHub Actions, Linux, Docker, Railway, AWS, GCP, pytest, JUnit, Maven, Arduino, Postman, Claude Code, Codex, Prompt Engineering, RAG
Concepts: Data Structures & Algorithms, OOP, Database Design, Unit/Integration Testing, CI/CD, Operating Systems